

1. 2.3 28

Solution: The range of $f(x) = e^x$ is only positive numbers (for instance, there is no solution to $0 = e^x$, so the function is not invertible with codomain all real numbers. However, with codomain positive real numbers, the function is invertible, since for any $y \in \mathbb{R}^+$, there exists $a = \ln(y) \in \mathbb{R}$ with $f(a) = e^{\ln(y)} = y$. Thus f is onto with the restricted domain. It is of course one-to-one (passes the horizontal line test), and hence invertible.

2. 2.3 34b

Solution: Let $f \circ g$ be one-to-one. Let $a, a' \in A$ and suppose that $g(a) = g(a')$. Thus $f(g(a)) = f(g(a'))$, since the inputs are the same. Since $f \circ g$ is one-to-one, it follows that $a = a'$, and hence we conclude that g , too, is one-to-one.

3. 2.3 34c

Solution: Suppose that $f \circ g$ is a bijection. Assume first that g is onto. Let $b, b' \in B$ and suppose that $f(b) = f(b')$. Since g is onto, there exist $a, a' \in A$ such that $g(a) = b$ and $g(a') = b'$. Consider $f(g(a)) = f(b)$ and $f(g(a')) = f(b')$. Thus $f(g(a)) = f(g(a'))$. Since $f \circ g$ is one-to-one, we have that $a = a'$, so $g(a) = g(a')$ and $b = b'$. Thus f is one-to-one.

For the reverse direction, suppose f is one-to-one. Let $b \in B$ and consider $f(b) \in C$. Since $f \circ g$ is onto, there exists $a \in A$ with $f(g(a)) = f(b)$. As f is one-to-one, it follows that $g(a) = b$, so g is onto.

4. 2.3 36

Solution: This problem is identical to 34b, above.

5. 2.3 42b

Solution: Let $f : A \rightarrow B$ and let $S \subseteq A$ and let $T \subseteq A$. Let $x \in f(S \cap T)$. Hence there exists $y \in S \cap T$ such that $f(y) = x$. Hence $y \in S$ and $y \in T$. Since $y \in S$, it follows that $x = f(y) \in f(S)$, and similarly $y \in T$ implies that $x \in f(T)$. Thus $x \in f(S) \cap f(T)$, proving that $f(S \cap T) \subseteq f(S) \cap f(T)$.

6. 46b

Solution: Let $f : A \rightarrow B$ and let $S, T \subseteq B$. Let $x \in f^{-1}(S \cap T)$. Thus $f(x) \in S \cap T$, so $f(x) \in S$ and $f(x) \in T$. Since $f(x) \in S$, it follows that $x \in f^{-1}(S)$, and analogously $x \in f^{-1}(T)$. Therefore $x \in f^{-1}(S) \cap f^{-1}(T)$, so $f^{-1}(S \cap T) \subseteq f^{-1}(S) \cap f^{-1}(T)$. The reverse direction is identical.

7. 2.5 4a

Solution: The set is countably infinite. For a one-to-one correspondence, consider the following piecewise function: $f : A \rightarrow \mathbb{Z}^+$ with (if $x = 3k + 1$ and $x > 0$) $f(x) = 4k + 1$; (if $x = 3k + 2$ and $x > 0$) $f(x) = 4k + 2$; (if $x = -3k + 1$ and $x < 0$) $f(x) = -4k + 3$; and (if $x = -3k + 2$ and $x < 0$) $f(x) = -4k$. Since each branch of the piecewise function is going to a different type of number mod 4, the function is one-to-one, and the function is onto because none of these are equal to 0 and all possible $k \geq 0$ are represented.

8. 2.5 4b

Solution: Again, this is countably infinite. You can use a piecewise function similar to that in 4a.

9. 2.5 4c

Solution: This set is countably infinite. For each real number, we need to know two things: the integer component before the decimal, and the positive integer number of ones after the decimal point. This set is thus in bijection with $\mathbb{Z} \times \mathbb{Z}^+$, which is countably infinite. For a one-to-one correspondence, use an array method like for the proof that $\mathbb{Z} \times \mathbb{Z}$ is countable.

10. 2.5 4d

Solution: This set is uncountable, because we have $\mathcal{P}(N)$ choices for decimal representations: each subset of N corresponds to the placement of 9s in the decimal sequence, with the remainder as ones. Since $\mathcal{P}(N)$ has uncountable cardinality, so too does this set.

11. 2.5 10

Solution: With finite difference: $A = (0, 1]$ and $B = (0, 1)$ have $A - B = \{1\}$.
With countable difference: $A = (0, 1) \cup N$ and $B = (0, 1)$ have $A - B = N$.
With uncountable difference: $A = \mathbb{R}$ and $B = (0, 1)$.

12. 2.5 12

Solution: Let $A \subseteq B$. Consider the function $f : A \rightarrow B$ with $f(a) = a$. This function is one-to-one, so $|A| \leq |B|$

13. 2.5 15

Solution: By problem 12, since $A \subseteq B$, it follows that $|A| \leq |B|$. Since A is uncountable, so too is B .

14. 2.5 30

Solution: Each quadratic equation is determined uniquely by a tuple of three integers, (a, b, c) . These are the elements of the set $Z \times Z \times Z$. Since the Cartesian product of countable sets is countable, it follows that there are countably many quadratic equations with integer coefficients. Each has at most two real solutions, so this is still countable.

15. 2.5 40

Solution: Suppose for a contradiction that f is an onto function from a set S to its power set $\mathcal{P}(S)$. Let $T = \{x \in S \mid x \notin f(x)\}$. Since f is onto, there exists $y \in S$ with $f(y) = T$. If $y \in T$, then by the definition of T , $y \notin f(y) = T$. If $y \notin T$, then by the definition of T , $y \in f(y) = T$. So y cannot be placed, for a contradiction. Hence no onto function exists.

16. 4.1 12

Solution: If x is odd, then x^2 is odd, so too is $x^2 + 2$, and the result is immediate. If x is even, then $2 \mid x$, so $4 \mid x^2$, and 4 does not divide $x^2 + 2$.

17. 4.1 18ace

Solution: a) $13 \cdot 11 \equiv 143 \equiv 10 \pmod{9}$.
c) $11 - 3 \equiv 8 \pmod{9}$.
e) $2(11^2) + 3(3^2) \equiv 242 + 27 \equiv 269 \equiv 3 \pmod{9}$.

18. 4.1 20

Solution: This is false: consider $78 \text{ div } 10$ and $24 \text{ div } 10$. We have $78 \text{ div } 10 + 24 \text{ div } 10 = 7 + 2 = 9$, but $(78 + 24) \text{ div } 10 = 102 \text{ div } 10 = 10$.

19. 4.1 26ab

Solution: a) 1 b) 4

20. 4.1 42

Solution: Let a, b, c, m be integers with $m \geq 2$ and $c \geq 0$ and $a \equiv b \pmod{m}$. Thus there exists $k \in \mathbb{Z}$ with $a = b + mk$. Thus $ac = bc + mkc$, so $ac \equiv bc \pmod{mc}$.

21. 4.1 43

Solution: a) Let $a = 2, b = 4, c = 3$, and $m = 6$. Then $ac \equiv 6 \equiv 0 \pmod{6}$ and $bc \equiv 12 \equiv 0 \pmod{6}$, but $a \not\equiv b \pmod{6}$.

b) Let $a = b = 3, c = 1, d = 6$, and $m = 5$. Then $a^b = 3^1 = 3$, but $c^d = 3^6 = 729 \equiv 4 \pmod{5}$.

22. 4.1 46

Solution: Let n be an odd positive integer, so we can write $n = 2k + 1$ for some nonnegative integer k . Thus $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1$. If k is even, then $4k^2$ and $4k$ are both divisible by 8, so $4k^2 + 4k + 1 \equiv 0 + 0 + 1 \pmod{8}$. Otherwise, k is odd, so we can write $k = 2l + 1$ for some nonnegative integer l . Thus $n^2 = 4k^2 + 4k + 1 = 4(2l + 1)^2 + 4(2l + 1) + 1 = 4(4l^2 + 4l + 1) + 8l + 4 + 1 = 16l^2 + 16l + 4 + 8l + 4 + 1 = 16l^2 + 24l + 9$; so we have $n^2 \equiv 16l^2 + 24l + 9 \equiv 0 + 0 + 1 \pmod{8}$. Hence for all odd n , n^2 is congruent to 1 $\pmod{8}$.

23. Lecture 6, Slide 6

Solution: Let a, b, c be integers with $a|b$ and $b|c$. Thus there exist integers m, n with $b = am$ and $c = bn$. By substitution, $c = amn$, so $a|c$.

24. Lecture 6, Slide 10

Solution: This is a fairly subtle/difficult proof.

Suppose that for some integers a and d , we can write $a = dq + r$ and $a = dq' + r'$, where $0 \leq r < d$ and $0 \leq r' < d$. Hence $dq + r = dq' + r'$, so $d(q - q') = r' - r$. Since r and r' are both between 0 and $d - 1$, it follows that $-d < r' - r < d$. Since d divides $d(q - q')$, it must also divide $r' - r$, but the only multiple of d within the bounds for $r' - r$ is 0. Hence $d(q - q') = r' - r = 0$, so $r' = r$ and (as d is nonzero) $q = q'$.

25. Lecture 6, Slide 12

Solution: Let $m|(b - a)$ for relevant integers m, b, a . Thus there exists k such that $mk = b - a$. This implies $a = b + m(-k)$, so $a \equiv b \pmod{m}$.